

Monte Carlo Integration: Recap

Mark M. Fredrickson (mfredric@umich.edu)

Module 1: Integrate — Session 1

Computational Methods in Statistics and Data Science (DATASCI 406)

Integrals as expectations

We wish to estimate $\int_a^b g(x) dx$. Suppose X is a random variable with:

- Support $[a, b]$ (where either $a \rightarrow -\infty$ or $b \rightarrow \infty$)
- Density $f(x)$ that is non-zero over $[a, b]$

then

$$\int_a^b g(x) dx = \int_a^b \frac{g(x)}{f(x)} f(x) dx = E(h(X)), \quad h(x) = g(x)/f(x)$$

Sample from X i.i.d. and estimate using $(1/n) \sum_{i=1}^n g(X_i)/f(X_i)$.

Special case: if the target is already $E(g(X))$ for an X we can sample, just sample and average — the density is baked into the sampler (`r` functions; `d` gives the density).

Probabilities are expectations of indicators

$$\mathbf{1}(s) = \begin{cases} 1 & : s \text{ is true} \\ 0 & : s \text{ is false} \end{cases} \quad \mathbb{E}(\mathbf{1}(X \in A)) = P(X \in A)$$

In R, indicators are free: `x <= 3`.

The empirical CDF is the sample mean of indicators:

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq t)$$

Why it works

Suppose $X_i \stackrel{\text{iid}}{\sim} F$, $E(g(X)) < \infty$, $E(g(X)^2) < \infty$. Then $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n g(X_i)$ is:

- **Unbiased** for any n : $E(\hat{\theta}_n) = E(g(X))$
- **Consistent** (weak law of large numbers): $\hat{\theta}_n \xrightarrow{P} E(g(X))$
- **Asymptotically Normal** (central limit theorem):

$$\frac{\hat{\theta}_n - \theta}{\sigma/\sqrt{n}} \xrightarrow{D} N(0, 1)$$

Always report a confidence interval

A Monte Carlo answer is an **estimate**, not an exact solution. By the CLT, an approximate $100(1 - \alpha)\%$ interval is

$$\hat{\theta} \pm t_{\alpha/2}(n - 1) \times \hat{s} / \sqrt{n}$$

- Continuous $h(X_i)$ values: `t.test(hX)$conf.int`
- Indicator values: `binom.test(sum(hits), n)$conf.int`

The width shrinks like $1/\sqrt{n}$: **each extra digit of precision costs 100 times more samples.**

Error rates and dimension

$$\text{Var}(\hat{\theta}_n) = \frac{\tau^2}{n}, \quad \tau^2 = \text{Var}(g(X)) \quad \Rightarrow \quad \text{error} = \tau n^{-1/2} = O(n^{-1/2})$$

- n sets the **rate**; τ sets the **constant**.
- Grid methods: $O(n^{-2/k})$ in k dimensions — ties Monte Carlo at $k = 4$, loses beyond.
- Monte Carlo's rate **does not depend on the dimension**: sample a random vector, apply g , average.